Notations:

- $\mathbb{k} = \mathbb{Q}(q,t)$
- $\Lambda = \Lambda_{\mathbb{k}}(X)$: algebra of symmetric functions in an infinite alphabet of variables $X = x_1, x_2, x_3, \dots$ with coefficients in $\mathbb{k}$
- $\omega : \Lambda \to \Lambda$: $\mathbb{k}$-algebra automorphism determined by $s_\lambda \mapsto s_{\lambda^*}$ (λ^* : conjugate of λ)
- For $f \in \Lambda$ and any expression A, $f[A]$ is the plethystic evaluation (see §2 in Pam)

  (* $f(x_1, x_2, \dots) = f[\underbrace{x_1 + x_2 + \cdots}_{A}]$ b/c $p_k[x_1 + x_2 + \cdots] = x_1^k + x_2^k + \cdots = p_k(x_1, x_2, \dots)$. We write $f[X] = f(X)$.) this $X = x_1 + x_2 + \cdots$ this $X = x_1, x_2, \dots$

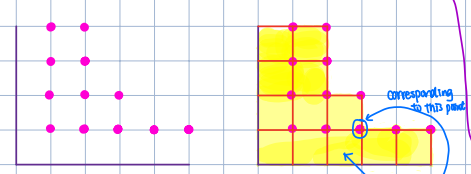
Recall that Λ is a Hopf algebra with coproduct $\Delta f = f[X+Y] \in \Lambda \otimes \Lambda = \Lambda(X) \otimes \Lambda(Y)$. We use X, Y to distinguish the tensor factors.

e.g.
- $p_n[X+Y] = p_n(X) + p_n(Y) = p_n(X) \cdot 1 + 1 \cdot p_n(Y)$
- $h_n[X+Y] = \sum_{j+k=n} h_j(X) h_k(Y)$
- $h_n[X-Y] = \sum_{j+k=n} h_j(X) h_k[-Y] = \sum_{j+k=n} h_j(X) \cdot (-1)^k e_k[Y] = \sum_{j+k=n} (-1)^k h_j(X) e_k(Y)$.
- $\Omega = 1 + \sum_{k \geq 1} h_k = \exp \sum_{k>0} \frac{p_k}{k}$. Note: $\Omega[a_1 + a_2 + \cdots - b_1 - b_2 - \cdots] = \frac{\prod_{i \geq 1}(1 - b_i)}{\prod_{i \geq 1}(1 - a_i)}$.
- $M = (1-q)(1-t)$, $\hat{M} = (1 - \frac{1}{qt}) M = (1 - \frac{1}{qt})(1-q)(1-t)$.

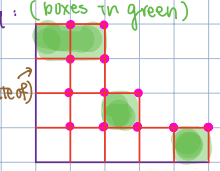
e.g. $\Omega[-\frac{\omega}{y}\hat{M}] = \Omega[-\frac{\omega}{y}(1-q)(1-t)(1-\frac{1}{qt})] = \Omega[q\frac{\omega}{y} + t\frac{\omega}{y} + \frac{1}{qt}\frac{\omega}{y} - \frac{1}{t}\frac{\omega}{y} - \frac{1}{q}\frac{\omega}{y} - qt\frac{\omega}{y}] = \frac{(1 - \frac{1}{q}\frac{\omega}{y})(1 - \frac{1}{t}\frac{\omega}{y})(1 - qt\frac{\omega}{y})}{(1 - q\frac{\omega}{y})(1 - t\frac{\omega}{y})(1 - \frac{1}{qt}\frac{\omega}{y})}$.

- (French style) Ferrers shape of $\lambda := \{(i,j) \mid 1 \leq j \leq \ell(\lambda),\ 1 \leq i \leq \lambda_j\}$
- lattice squares (boxes) of λ = boxes with NE corner in Ferrers shape of λ
- skew diagram : difference $\nu = \lambda/\mu$ of partition diagrams $\mu \subseteq \lambda$ or any translate of such a diagram
- For a box $a = (i,j)$, let south$(a) = (i, j-1)$ denote the box immediately south of a
  north$(a) = (i, j+1)$ denote the box immediately north of a
  east$(a) = (i+1, j)$ denote the box immediately east of a
  west$(a) = (i-1, j)$ denote the box immediately west of a
- content of a box $a = (i,j)$ = $i - j$. (j^{th} row from left; i^{th} column from bottom)

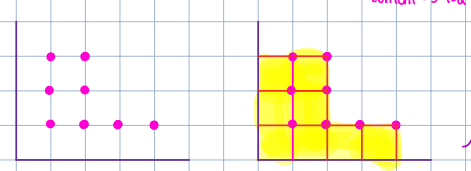
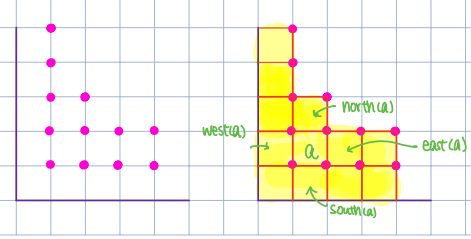
Example: $\lambda = (5,3,2,2,0,0,0)$. We have
- $|\lambda| = 12$, $\ell(\lambda) = 4$
- Ferrers shape :
- Boxes :



$\therefore$ λ/μ : (boxes in green)



$\mu = (4,2,2,0,0,0,0)$
- Ferrers shape :
- Boxes :



$\lambda^* = (4,4,2,1,1)$
- Ferrers shape :
- Boxes :